

JAESEOK HWANG

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Ph.D. in Applied Economics | Decision-Risk Analytics · Interpretable ML · Production Data Pipelines

RESEARCH EXPERIENCE

University of Illinois Urbana-Champaign

Sep 2020 – Present

Research Assistant, DIFM Project (USDA-funded)

- Built end-to-end data pipelines to automate decision-support workflows from API/data queries and SQL processing through analysis and producer-facing reporting for a USDA-funded initiative with approximately **\$70M** in total project scale.
- Collaborated with Oracle Cloud engineers and lead economic analysts to support implementation and data operations for a 16-university consortium.
- Used R (mlr3, caret, xgboost) and Python (scikit-learn, NumPy, GeoPandas) to estimate dynamic profit structures under price and environmental uncertainty.

DISSERTATION PAPERS (Research Projects)

Information Transferability and Economic Risk Without Structural Validation

- Developed a validation framework for non-stationary environments using monotonicity constraints and leave-one-location-out stress tests.
- Mapped technical prediction error to **economic loss proxies**, creating a practical go/no-go rule for model deployment in high-stakes farm management settings.

Project II: Decomposition of Economic Risk (Ex-ante vs. Ex-post)

- Built Bayesian hierarchical and nonlinear models to decompose **profit-at-risk** into decision-time uncertainty versus realized environmental shocks.
- Applied partial pooling and Bayesian shrinkage to stabilize inference in sparse panels and improve reliability of recommendations in high-variance regions.

Project III: Bayesian Learning & Value of Information (VOI)

- Developed a simulation framework to quantify the **value of learning** from new experiments and identify the convergence path of decision quality over time.
- Showed how favorable and unfavorable weather realizations shift posterior learning and **economic regret**, informing the return to additional data collection.

METHODS & TOOLS

- **Economic Methods:** Structural econometrics, causal inference, Bayesian hierarchical modeling, panel data analysis, Monte Carlo simulation.
- **Decision Analytics:** Decision-loss decomposition, stress testing, value of information (VOI), profit-at-risk analysis.
- **Programming:** Python, R, SQL, Git, geospatial data workflows.

EDUCATION

University of Illinois at Urbana-Champaign

May 2026 (expected)

Ph.D. Agricultural and Applied Economics (Focus: Econometrics, Optimization, Price Analysis)

University of Illinois Urbana-Champaign

May 2020

M.A. Agricultural and Applied Economics

Sogang University

Aug. 2013

B.S. Economics